> **Faculty of Information Technology (FIT) – Ho Chi Minh City University of Science (HCMUS)**
> CS423 / CSC13003 – Software Testing (AI-augmented · 2026)
> *AI POLICY · TEMPLATES — 2026 v1.0*

AI Audit Report — 5-section Template per Artifact

Mandatory appendix for every AI-assisted homework (HW#01–HW#06, and Seminar).

Adapted from Med Kharbach, PhD (2026) — AI Use Policy Templates for Higher Education. CC BY-NC-SA 4.0. This adaptation is prepared for FIT@HCMUS – CS423 / CSC15003 Software Testing course.

1. Student Information

Student name (printed): Hà Bảo Ngọc
Student ID: 23127300
Class / Cohort: 23KTPM3
Assignment ID (e.g., HW#00, HW#02): HW01
Assignment date: 07/06/2026
AI tool(s) used: Claude, ChatGPT, Gemini

2. Instructions (read before filling)

> [!IMPORTANT]
> - Add one row per AI-generated artifact (test case, script, checklist, OpenAPI spec, JMeter plan, etc.).
> - Paste the verbatim prompt — DO NOT paraphrase.
> - Paste the verbatim AI output (or include a labelled screenshot in the report).
> - Tag the verdict: **VALID** / **INVALID** / **INCOMPLETE**.
> - Reasoning must cite a course slide, ISTQB section, or technical RFC.
> - Show the corrected artifact with the change highlighted.
> - Sample rows are in italic — replace them before submission.

3. Audit Table — one row per artifact

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
▶ **Artifact #1 — Test Cases (Pedestal Fan)**	▶ 15 test cases generated...	**VALID**	Baseline test cases generated by AI are expected to be generic and cover standard functional paths. Per prompt.txt NOTE: *"the test cases is pretty safe and typical, will not find the defects, added some edge cases to find the defects"* — this was the student's intended workflow, not an AI shortcoming. ISTQB FL §4.1 confirms test cases should be based on product specifications; the AI correctly used the pedestal fan as the specification baseline. ISTQB FL §4.4 and §4.5 note that boundary conditions and unusual sequences (e.g., concurrent state changes) are the analyst's responsibility to add post-generation.	**(1)** Added TC16 (oscillation interruption recovery) and TC17 (speed change during oscillation) as concurrent-state edge cases. <br**(2)** TC10 (height adjustment) noted as inapplicable to Mitsubishi LV16-RM — this TC was identified post-audit and corrected in HW01_Report.md. AI-generated baseline accepted as-is.
▶ **Artifact #2 — ISTQB QA/QC Mindmap**	▶ A markdown-structured mindmap...	**INCOMPLETE**	**(1)** The mindmap placed "Performance Testing" under "Functional Testing." ISTQB FL §2.1 classifies performance testing as a **non-functional** testing type — this is a category error. **(2)** "Static Testing" was listed as a QA task. Per ISTQB FL §2.1, static testing (reviews, static analysis) is a testing technique applicable by both QA and QC, not exclusive to QA. **(3)** The mindmap omitted "Test Automation" as a	**(1)** Moved "Performance Testing" out of Functional Testing into its correct Non-Functional Testing branch. **(2)** Re-labeled "Static Testing" as a cross-functional technique, not exclusive to QA. **(3)** Added "Test Automation" as a dedicated branch under QA, covering automation strategy,

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			separate branch despite it being a key QA activity. **(4)** Per prompt.txt NOTE: *"the mindmap quality is not very good visual-wise and content-wise, explained in the markdown."* The visual quality issue (unbalanced branch depth, sparse tool section, duplicate nodes) was also flagged by the student — the audit addresses both the taxonomy errors and the visual layout issues above.	framework selection, and CI/CD integration. **(4)** Improved visual balance by expanding sparse tool branches (added LoadRunner, Gatling, Burp Suite, OWASP ZAP), consolidating duplicate Reporting nodes, and removing redundant Risk Management entries. Corrected mindmap saved in `QA_QC_Mindmap.md`.
▶ **Artifact #3 — Defect Explanations (20 Software Defects)**	▶ Explanations for all 20 defects...	**INCOMPLETE**	Per prompt.txt NOTE: *"prompt 1: overall good but there are some biases/hallucination in the explanation, details in the report."* This means the AI output itself was silent or imprecise on the specific hallucination details — the student filled those in manually based on authoritative sources (CISA, OWASP, CVE databases), not from the AI response. This distinction is important: the AI generated vague or incomplete descriptions, and the student independently researched and added the corrected technical and legal details. Specific failures identified: **(1)** Air Canada chatbot: called	**(1)** Student independently cross-referenced each AI explanation against authoritative sources (CISA advisories, OWASP, CVE databases, news articles) and added the corrected details manually (as noted: *"details in the report"*). **(2)** For Air Canada: added note questioning whether it was truly an LLM hallucination vs. a rule-based system error. **(3)** For EEOC: added the "Vendor Defense

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			it an "LLM hallucination" without confirming the bot was actually LLM-based — it may have been rule-based. **(2)** EEOC hiring bias: failed to note the "Vendor Defense" is dead — employers are strictly liable for third-party AI tools per post-2023 EEOC guidance. **(3)** MOVEit: skipped the LEMURLOOT web shell deployment — the actual data-theft mechanism. **(4)** CrowdStrike: lacked technical root cause (out-of-bounds memory read in Channel File 291, kernel mode). **(5)** XZ Utils: missed the systemd dependency trick and pre-authentication RCE nature. ISTQB FL §5.2 requires test analysts to verify information from multiple sources — AI summaries alone are insufficient for accuracy.	is dead" legal nuance in the AI Bias column. **(4)** For CrowdStrike: added the Channel File 291 / C-2 / out-of-bounds memory read technical detail. **(5)** For XZ Utils: added the systemd dependency trick and pre-auth RCE classification. All corrections documented in the AI Bias/Hallucination Instance column in `HW01_Report.md` Requirement 2 section.
▶ **Artifact #4 — HW01 Report Structure and Req 1 (Job Market)**	▶ Full markdown report structure...	**VALID**	**(1)** Prompt #1 of this artifact initially omitted the Student Info section and appendix sections, and added grading point allocations (e.g., "40 pts"). Per prompt.txt NOTE on prompt #1: *"done ok but needs fixing in prompt 2."*	No further fix required. The follow-up prompt (prompt #2) produced an acceptable result per the student's own assessment. Minor cleanup of redundant

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
			(2) Prompt #2 (the follow-up correction) successfully applied all fixes: Student Info added, grading metadata removed, screenshots converted to markdown image syntax, appendix sections included. Per prompt.txt NOTE on prompt #2: *"result is ok no need manual fix."* **(3)** Note: The student manually prepared `req_1_materials.txt` and placed screenshots in `screenshots/requirement_1` before these prompts — AI generated the draft structure and job content from human-sourced data inputs. ISTQB FL §5.3: Test documentation should follow the agreed format — after the follow-up correction, the output met the required format. The audit evaluates the corrected output, not the initial draft.	heading levels done by student.
▶ **Artifact #5 — HW01 Report Req 2 (20 Software Defects)**	▶ Requirement 2 section added...	**INCOMPLETE**	**(1)** Gemini did not automatically search for missing links — when prompted with req_2_links.txt separately (prompt #4), it still pasted placeholder links instead of resolving the actual URLs (e.g., "Fortinet CVE-2022-42475 Security Advisory" was not resolved to the actual advisory URL). **(2)** Per prompt.txt NOTE:	**(1)** After Gemini failed to resolve links automatically, student manually looked up each source link from req_2_links.txt and replaced placeholder text with verified URLs. **(2)** Verified links by testing each URL resolves correctly

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
			"result is not very good, the materials show some as not link but required to search and paste the real link in, it just pasted the requirement not searching it itself." Note: The 20-defect content framework was first explained by ChatGPT (Artifact #3); Gemini used `req_2_materials.txt` as the source of truth and built the report section from it. ISTQB FL §5.2: Information used as a test basis must be verifiable and traceable to authoritative sources.	before adding to report.
▶ **Artifact #6 — Prompt Log Outline**	▶ A markdown outline template...	**VALID**	The outline correctly structured the prompt log with: chronological ordering, AI tool identification, verbatim prompt fields, screenshot reference fields, and a notes section for student commentary. ISTQB FL §5.3: Good test documentation requires traceability — the prompt log structure supports tracing AI outputs back to inputs, which is the basis of the AI Audit Report. Per prompt.txt NOTE: "result is ok, there are some redundant lines but can easily be manually removed." No substantive errors requiring rejection.	Minor cleanup: removed a few redundant heading levels and filler lines. Otherwise accepted as-is.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
▶ **Artifact #7 — Homework Overview and Step-by-Step Guidance (Exploratory)**	▶ A high-level homework overview...	**VALID**	This was an exploratory/advisory prompt — Claude provided general guidance on homework structure, AI document requirements, and workflow. No specific technical artifacts were generated that required verification. Per prompt.txt NOTE: "mostly homework exploratory questions, so doesn't need manual fixing, and helped me to get the overall requirements and what to do." ISTQB FL §1.2: Testing activities benefit from proper planning and guidance — this output served a planning function, not a test artifact production function. No defect-inserting errors were identified.	No fix required.
▶ **Artifact #8 — HW01 Report Req 3 (Physical Product Test Cases)**	▶ Requirement 3 section drafted...	**INCOMPLETE**	**(1)** Gemini initially generated TC10 (height adjustment) which is inapplicable to the Mitsubishi LV16-RM (fixed-post-height model). ISTQB FL §4.1 requires test cases to be based on actual product specifications. **(2)** The AI-generated test cases covered standard functional paths but omitted concurrent-state edge cases (e.g., speed change during oscillation, oscillation interruption recovery). Per ISTQB FL §4.4 and §4.5,	**(1)** Removed TC10 (height adjustment) — inapplicable to fixed-post Mitsubishi LV16-RM. **(2)** Added FAN-TC007 (oscillation interruption recovery) and FAN-TC015 (speed change during oscillation) as concurrent-state edge cases. Both revealed actual product defects.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
			boundary conditions and unusual sequences require analyst judgment beyond AI consensus-level output.	**(3)** Added serial number `90402931` via follow-up prompt (Artifact #9).
▶ **Artifact #9 — Serial Number Addition**	▶ Serial number added to report...	**VALID**	Simple factual correction — the student provided the exact serial number and Gemini correctly inserted it into the device information section. No technical judgment required; output is verifiable by inspection. ISTQB FL §5.3: Test documentation must be accurate and complete — the serial number is a basic completeness requirement.	No fix required.
▶ **Artifact #10 — Prompt Log Final Update**	▶ Full conversation updated to prompt_log.md...	**VALID**	Gemini correctly compiled the full conversation into the prompt_log.md format with accurate timestamps, prompt text, and screenshot reference fields. Per prompt.txt NOTE: "result is ok, there are some redundant lines but can easily be manually removed." No substantive errors requiring rejection. ISTQB FL §5.3: Good test documentation requires traceability — the prompt log supports tracing AI outputs back to inputs for the AI Audit Report.	Minor cleanup: removed redundant filler lines. Otherwise accepted as-is.
▶ **Artifact #11 — Defect List Refinement**	▶ Refined list of 20 defect sources...	**VALID**	Claude served as a pre-screening advisor for the 20-defect list, helping the student identify which	No fix required. The student then passed the finalized list to

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
and Finalization			sources were well-documented vs. niche/ambiguous. Per prompt.txt NOTE: *"prompt 16: good sources, but some require manual searching for source."* (Note: prompt.txt has two entries labeled "prompt 16" — see below.) The screening function is a valid AI use case under brainstorming/outlining. ISTQB FL §1.2: Proper planning and source selection are foundational to testing — Claude's advisory role here is analogous to a peer review of the test basis.	ChatGPT (Artifact #3) for full explanations.
▶ **Artifact #12 — Remaining Claude Prompts (Exploratory / Advisory)**	▶ Advisory responses and conceptual clarifications...	**VALID**	These 11 prompts (Prompts 2–12) were exploratory and advisory in nature — they answered homework mechanics, clarified requirements, and discussed strategy. Per prompt.txt NOTE: *"mostly homework exploratory questions, so doesn't need manual fixing."* No specific technical artifacts were generated that could contain defect-inserting errors. The outputs served a planning and comprehension function. ISTQB FL §1.2: Testing activities benefit from proper planning and guidance — these outputs served a planning function, not a test artifact production function.	No fix required. These were informational/advisory prompts only.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
			No substantive errors were identified.	
▶ **Artifact #13 — Claude Prompt #17 (Final Prompt Log Compilation)**	▶ A compiled prompt log...	**INCOMPLETE**	Per prompt.txt NOTE: *"prompt 18: good prompt log, but lacks some messages (i copied myself), some long messages got shortened (i fix by copy) and needs manual copy the timestamp."* (Note: prompt.txt labels this as "prompt 18" in the notes section, but it corresponds to the 18th Claude prompt chronologically.) **(1)** Some messages from the Claude conversation were not captured — the student had to manually copy them in. **(2)** Some long messages were truncated or shortened by the AI. **(3)** Timestamps needed to be manually copied by the student. ISTQB FL §5.3: Good test documentation requires completeness and traceability — missing or truncated prompts undermine the audit trail.	**(1)** Student manually copied missing messages into prompt_log.md. **(2)** Student manually expanded truncated messages with full text. **(3)** Student manually added accurate timestamps where the AI output was incomplete.
▶ **Artifact #14 — Gemini Prompt #6 (Partial Report Compilation)**	▶ Partial report sections compiled...	**VALID**	Per prompt.txt NOTE: *"prompt 6: result is ok."* The student explicitly confirmed this output was satisfactory. Gemini correctly followed the instruction to skip Req 3 and the prompt log, and to include the AI Critique and Self-	No fix required.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
			Assessment. No corrections needed. ISTQB FL §5.3: Test documentation should follow the agreed format — the output matched the student's instructions and was accepted without further correction.	

4. Summary of AI Accuracy

Metric	Count	Percentage
Total AI-generated artifacts audited	14	100%
VALID (correct, accepted as-is)	7	50%
INVALID (wrong; rejected)	0	0%
INCOMPLETE (acceptable after edits)	7	50%

5. Conclusion — When should AI be used (or not)?

AI performed best as a **structural accelerator** — drafting report outlines, generating initial test case templates, explaining broad concepts, and compiling prompt logs. Both ChatGPT and Gemini produced usable starting points for the 10-job market analysis and the 20-defect descriptions that would have taken significantly longer to draft from scratch. Gemini's prompt log outline and partial report compilation required minimal correction.

AI performed worst when **specialized domain precision was required**. The 20-defect explanations consistently omitted critical technical details (e.g., LEMURLOOT web shell in MOVEit, Channel File 291 in CrowdStrike, pre-auth RCE classification for XZ Utils) and legal nuances (employer strict liability for AI hiring tools). ChatGPT's test cases included a test case inapplicable to the actual product (height adjustment), and the QA/QC mindmap misclassified performance testing under functional testing. These are not random errors — they reflect AI's tendency to produce consensus-level summaries that prioritize plausibility over precision.

AI's exploratory/advisory prompts (11 of Claude's 18 prompts) served a planning function only — they answered homework mechanics, clarified requirements, and discussed strategy. None of these produced defect-inserting technical artifacts, per the student's own assessment in prompt.txt: *"mostly homework exploratory questions, so doesn't need manual fixing."*

Notable pattern: Multiple Gemini prompts followed a 2-step correct-and-retry pattern — the initial output was flawed, the student requested corrections, and the follow-up output was accepted as satisfactory. This was documented per prompt.txt notes (e.g., *"prompt 1 done ok but needs fixing in prompt 2"*, *"prompt 2 result is ok no need manual fix"*).

Recommendation: Use AI for **bootstrapping drafts and brainstorming structures**, then apply rigorous human verification using authoritative sources (CVE databases, official advisories, ISTQB syllabus). Never accept AI-generated technical or legal content at face value. The most reliable workflow is AI First → Human Review → AI Audit Report → Git commit.

6. Mandatory Disclosure (paste verbatim)

> *"The report structure, defect explanations, QA/QC mindmap, and initial test cases were initially generated by Gemini, ChatGPT, and Claude; I reviewed and modified the job market analyses and software defects sections, added specific edge cases to the physical device test cases; the physical device testing execution, videos, device photos, and dated screenshots were produced entirely by me. The detailed AI Audit Report is attached as the [AI-02] document (with the prompt log attached as Appendix A). I confirm I did not use AI to generate any artifact listed in the prohibited category below."*

Signature

Student name (printed): Hà Bảo Ngọc
Student ID: 23127300
Class / Cohort: 23KTPM3
Course: CS423 / CSC13003 – Software Testing
Instructor: MSc. Trần Thị Bích Hạnh, MSc. Hồ Tuấn Thanh
Date: 07/06/2026
Signature: *Hà Bảo Ngọc*

References

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